

Wilson loop dependence on the contour shape

H. A. Falomir

Laboratorio de Física Teórica, Universidad Nacional de La Plata, Argentina

R. E. Gamboa Saravi*

*Center for Theoretical Physics, Laboratory for Nuclear Science and Department of Physics,
Massachusetts Institute of Technology, Cambridge, Massachusetts 02139*

F. A. Schaposnik*

*Laboratoire de Physique Théorique et Hautes Energies,[†]
Université Pierre et Marie Curie, Paris, France*

(Received 1 September 1981; revised manuscript received 29 September 1981)

By computing exactly the Wilson loop expectation value for two-dimensional quantum electrodynamics, we show that the nonleading terms depend drastically on the shape of the contour.

I. INTRODUCTION

One of the basic tools in the study of quantum chromodynamics is the Wilson loop expectation value

$$W[C] = \left\langle \text{Tr} P \exp \left[\oint_C ig A_\mu dx^\mu \right] \right\rangle.$$

As is well-known, $W[C]$ provides a criterion for confinement, since it can be related to the binding energy of a quark-antiquark pair.¹ Furthermore, it can be used as a gauge-invariant dynamical variable on its own² in a search for an evolution equation whose solution should yield to confinement and, more generally, to a deeper understanding of hadron physics. In this last approach $W[C]$ replaces the traditional local variables and provides a link between QCD and dual string models. From this point of view, it is important to find solutions of the stringlike equations, involving variational derivatives with respect to C , proposed for $W[C]$.

In this paper we present the computation of $W[C]$ for a very simple gauge theory: the Schwinger model³ (two-dimensional quantum electrodynamics with massless quarks),⁴ hoping that our results, which are interesting by themselves, might also be helpful in the study of the interquark potential of more realistic theories, as well as in the search for solutions of the loop equations.

Our main conclusion concerns the highly nontrivial dependence of the Wilson loop on the shape of the contour. This behavior, already signaled by Abud, Bollini, and Giambiagi,⁵ and by Shifman⁶ for small contours in a perturbative approach to

QCD, has interesting implications on the analysis of corrections to $W[C]$ due to long-wavelength fluctuations of the string.⁷ In fact, we will show that even the first nonleading term in the Wilson loop expectation value depends drastically on the shape of the contour used in its computation. At the same time, this dependence must be accounted in the search for an ansatz for $W[C]$ which should solve the loop equations.

Even though the quark-antiquark potential must be evaluated by using a rectangular loop it is known that sometimes the contour has to be smooth in order to avoid singularities.⁷ But we shall show that, in the model we shall consider, nonleading terms depend sensitively on the contour shape used to round the corners.

We want to stress at this point that we shall consider the Wilson loop for the complete two-dimensional QED theory (i.e., when gauge fields and massless fermions are present). This is so for the following reasons. (i) In order to use $W[C]$ as the relevant dynamical variable in the loop-space approach to QCD, it is desirable to include fermions from the beginning. (ii) In $d=2$ dimensions, the Wilson loop expectation value for pure gauge theories gives a trivial area law which reflects the kinematical confinement characteristic of one spatial dimension. It is only when fermions are included that the mechanism for quark trapping (via the screening process) can be discovered.^{3,8,9}

Our study will be based on the computation of the *exact* Wilson loop expectation value for differently shaped contours.

II. LOOP COMPUTATION

We begin from the two-dimensional (Euclidean) Lagrangian

$$\mathcal{L} = -\bar{\psi} D \psi + \frac{1}{4} F_{\mu\nu} F_{\mu\nu} + \mathcal{L}_{\text{gauge fixing}}, \quad (1)$$

where $D = \gamma_\mu D_\mu = \gamma_\mu (i\partial_\mu + eA_\mu)$, and the γ matrices we are using satisfy $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$ and $\gamma_\mu \gamma_5 = i\epsilon_{\mu\nu} \gamma_\nu$ with $\gamma_5 = i\gamma_0 \gamma_1$ and $\epsilon_{01} = -\epsilon_{10} = 1$. We shall work in the Lorentz gauge $\partial_\mu A_\mu = 0$ by taking

$$\mathcal{L}_{\text{gauge fixing}} = \frac{1}{2\alpha} (\partial_\mu A_\mu)^2.$$

We shall consider the loop average

$$\begin{aligned} W[c] &= \left\langle \exp \left[iQ \oint_C A_\mu dx_\mu \right] \right\rangle_{A, \bar{\psi}, \psi} \\ &= N \int \mathcal{D}A_\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[- \int \mathcal{L} d^2x \right] \\ &\quad \times \exp \left[iQ \oint_C A_\mu dx_\mu \right], \end{aligned} \quad (2)$$

where C is a closed curve defined by $x_\mu = x_\mu(t)$, and N is a normalized factor.

It is convenient to change the fermionic variables by means of the γ_5 transformation

$$W[c] = N \int \mathcal{D}A_\mu \mathcal{D}\bar{\chi} \mathcal{D}\chi \exp(-S_{\text{eff}}[A, \bar{\chi}, \chi]) \exp \left[iQ \oint_C A_\mu dx_\mu \right], \quad (6)$$

where

$$S_{\text{eff}}[A, \bar{\chi}, \chi] = \int \left[-i\bar{\chi} \partial \chi + \frac{1}{4} F_{\mu\nu} F_{\mu\nu} + \frac{e^2}{2\pi} A_\mu A_\mu + \frac{1}{2\alpha} (\partial_\mu A_\mu)^2 \right] d^2x. \quad (7)$$

The fermionic path integral is trivial, and Eq. (6) becomes

$$W[c] = N_A \int \mathcal{D}A_\mu \exp \left\{ - \int \left[-\frac{1}{2} A_\mu \partial^2 A_\mu + \frac{1}{2} \left(1 - \frac{1}{\alpha} \right) A_\mu \partial_\mu \partial_\nu A_\nu + \frac{\mu^2}{2} A_\mu A_\mu + J_\mu A_\mu \right] d^2x \right\}, \quad (8)$$

where $\mu = e/\sqrt{\pi}$ is the “photon” mass arising from the Schwinger mechanism,³ and the current is

$$J_\mu(\vec{x}) = iQ \int \delta^{(2)}(\vec{x} - \vec{x}(t)) \frac{dx_\mu(t)}{dt} dt. \quad (9)$$

Being Gaussian, the path integral in Eq. (8) can be computed exactly, giving

$$\begin{aligned} W[c] &= \exp \left[\frac{1}{2} \int J_\mu(\vec{x}) \Delta_{\mu\nu}(\vec{x} - \vec{y}) J_\nu(\vec{y}) \right. \\ &\quad \left. \times d^2x d^2y \right]. \end{aligned} \quad (10)$$

In momentum space, we can write

$$\begin{aligned} \psi(x) &= e^{\gamma_5 \phi(x)} \chi(x), \\ \bar{\psi}(x) &= \bar{\chi}(x) e^{\gamma_5 \phi(x)}, \end{aligned} \quad (3)$$

where ϕ satisfies the relation

$$A_\mu = -\frac{1}{e} \epsilon_{\mu\nu} \partial_\nu \phi, \quad (4)$$

a choice which is always possible in the Lorentz gauge.

This transformation decouples fermions from the gauge field in the Lagrangian (1). On the other hand, the path-integral measure is not invariant under the chiral transformation,¹⁰ and it has been proved^{11,9} that it is just this Jacobian factor, J , which gives rise to the “photon” mass term in the path-integral formulation of the model,

$$J = \exp \left[-\frac{e^2}{2\pi} \int A_\mu A_\mu d^2x \right], \quad (5)$$

making apparent the close relation between the axial-vector-current anomaly and the spontaneous generation of the “photon” mass.¹²

Therefore, Eq. (2) can be written as follows:

$$\ln W[c] = \frac{1}{8\pi^2} \int d^2k j_\mu(-\vec{k}) \frac{1}{k^2 + \mu^2} j_\mu(\vec{k}), \quad (11)$$

since, as usual, the α -dependent part of the propagator does not contribute owing to current conservation.

First let us choose a rectangular path, shown in Fig. 1, of sides T and d as curve c . Then the current becomes

$$\begin{aligned} j_0(\vec{k}) &= -4 \frac{Q}{k_0} \sin(k_0 T/2) \sin(k_1 d/2), \\ j_1(\vec{k}) &= 4 \frac{Q}{k_1} \sin(k_0 T/2) \sin(k_1 d/2), \end{aligned} \quad (12)$$

showing clearly its conservation ($k_\mu j_\mu = 0$).

Inserting Eq. (12) into Eq. (11) we obtain

$$\begin{aligned}\ln W_{\text{rect}}(T, d) &= -\frac{8Q^2}{\pi^2} \int_0^\infty dk_0 \int_0^\infty dk_1 \frac{\sin^2(k_0 T/2)}{k_0^2} \frac{k^2}{k^2 + \mu^2} \frac{\sin^2(k_1 d/2)}{k_1^2} \\ &= -\frac{2Q^2}{\pi} \int_0^\infty dk_1 \frac{\sin^2(k_1 d/2)}{k_1^2 + \mu^2} \left[T + \mu^2 \frac{1 - \exp[-T(k_1^2 + \mu^2)^{1/2}]}{k_1^2(k_1^2 + \mu^2)^{1/2}} \right].\end{aligned}\quad (13)$$

As usual, we define the effective potential between the charges $\pm Q$ as

$$V_{\text{rect}}(d) \equiv -\lim_{T \rightarrow \infty} \left[\frac{1}{T} \ln W(T, d) \right]. \quad (14)$$

Then we finally get

$$V_{\text{rect}}(d) = \frac{Q^2}{2\mu} (1 - e^{-\mu d}), \quad (15)$$

which shows that for short distances ($d \ll 1/\mu$) the potential behaves as expected¹³: $V \sim (Q^2/2)d$, while the limit for large distances makes the screening apparent.

This result was to be expected (and in fact, was advanced by Rothe *et al.*¹⁴ by studying a classical-mechanics analog of the Schwinger model): the potential corresponds to the immediate onset of the screening process, which occurs for any value of Q .

Now, if instead of a rectangle we consider a cir-

cular contour, the current becomes

$$j_\mu(\vec{k}) = -2\pi QR \epsilon_{\mu\nu} \frac{k_\nu}{k} J_1(kR) \quad (16)$$

as can be shown by explicitly computing the Fourier transform of Eq. (9) for a circle of radius R . Now, using this result in Eq. (11) we get

$$\begin{aligned}\ln W_{\text{circ}}(R) &= -Q^2 R^2 \pi \int_0^\infty \frac{k dk}{k^2 + \mu^2} [J_1(kR)]^2 \\ &= -Q^2 R^2 \pi I_1(\mu R) K_1(\mu R),\end{aligned}\quad (17)$$

where I_1 and K_1 are the modified Bessel functions of order 1 and R is the radius of a circle of area $\pi R^2 = Td$. This last expression shows a manifestly different behavior from that governed by Eq. (3). In particular, if by analogy with the rectangular case we define the potential between charges $\pm Q$ as¹⁵

$$V_{\text{circ}}(d) \equiv -\lim_{T \rightarrow \infty} \left[\frac{1}{T} \ln W_{\text{circ}}(R) \right], \quad (18)$$

then by combining Eqs. (17) and (18) we get

$$V_{\text{circ}}(d) = Q^2 d I_1(\mu d) K_1(\mu d). \quad (19)$$

The two potentials (V_{rect} and V_{circ}) coincide up to the leading term when $d \rightarrow 0$ and $d \rightarrow \infty$:

$$V_{\text{rect}}(d) \sim V_{\text{circ}}(d) \sim \frac{Q^2}{2} d, \quad d \ll 1/\mu, \quad (20a)$$

$$V_{\text{rect}}(d) \sim V_{\text{circ}}(d) \sim \frac{Q^2}{2\mu}, \quad d \gg 1/\mu. \quad (20b)$$

However, if we look at the large-distances expansion of $V_{\text{circ}}(d)$ we get the expression

$$V_{\text{circ}}(d) \sim \frac{Q^2}{2\mu} \left[1 - \frac{3}{8} \frac{1}{(\mu d)^2} - \frac{45}{128} \frac{1}{(\mu d)^4} - \dots \right] \quad (21)$$

showing a long-range force between pairs which should indicate the presence of a sort of van der Waals force between "mesons" (screened charges) at long distances.

This simple example shows that one must be

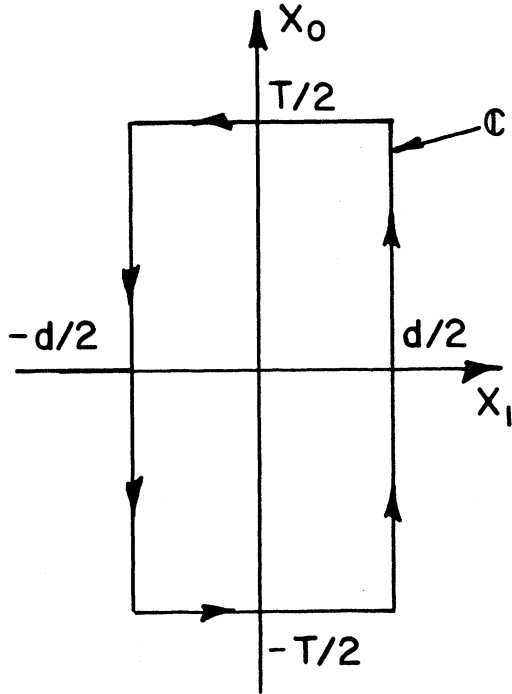


FIG. 1. Curve chosen to compute the loop average W_{rect} .

aware of the shape dependence of $W[C]$ when drawing physical conclusions from Wilson loops corresponding to particular contours. In particular, a circular loop, already considered in the literature in connection with instanton solutions,¹⁶ construction of loop equations¹⁷ as well as in the string approximation to the Wilson loop correlation function,¹⁸ seems to be very pathological in the sense that it leads to long-range forces for the interquark potential, which is a purely fictitious result for our model.

One can argue that this last result is due to the

$$j_\mu(\vec{k}) = -2\pi Qab\epsilon_{\mu\nu}k_\nu \frac{J_1((k_0^2a^2 + k_1^2b^2)^{1/2})}{(k_0^2a^2 + k_1^2b^2)^{1/2}}, \quad (22)$$

where a, b are the parameters of the ellipse ($a > b$). By inserting Eq. (22) in Eq. (11) we get

$$\ln W_{\text{ellip}}(a, b) = -2Q^2a^2b^2 \int_a^b \frac{d\rho}{\rho(a^2 - \rho^2)^{1/2}(\rho^2 - b^2)^{1/2}} I_1(\mu\rho) K_1(\mu\rho). \quad (23)$$

By defining the potential¹⁹

$$V_{\text{ellip}}(d) \equiv - \lim_{T \rightarrow \infty} \left[\frac{1}{T} \ln W_{\text{ellip}}(T/2, 2d/\pi) \right] \quad (24)$$

we find

$$V_{\text{ellip}}(d) = \frac{Q^2}{4\mu} \pi [I_1(4\mu d/\pi) - \mathbf{L}_1(4\mu d/\pi)], \quad (25)$$

where $\mathbf{L}_1$ is the modified Struve function of order 1. This potential, V_{ellip} , also satisfies relations (20), and its expansion for large distances

$$V_{\text{ellip}}(d) \sim \frac{Q^2}{2\mu} \left[1 - \frac{\pi^2}{16} \frac{1}{(\mu d)^2} - \frac{3\pi^4}{256} \frac{1}{(\mu d)^4} - \dots \right] \quad (26)$$

shows that, even in this unambiguously defined potential, long-range interactions are still present.

At this point we want to show our results for two dimensional QED easily generalize to massless diagonal two-dimensional QCD (DQCD₂). Recently it was found^{20,21,9} that the latter, where the gauge field is the maximal Abelian subgroup—the torus—of $SU(N)$, is also an exactly soluble theory, and that it shares the main features of QED₂; and at the same time it is thought it could play an im-

portant role in the solution of the complete QCD₂ theory. For this theory the Lagrangian is

$$\mathcal{L} = -\bar{\psi} D \psi + \frac{1}{4} F_{\mu\nu}^a F_{\mu\nu}^a + \mathcal{L}_{\text{gauge fixing}}, \quad (27)$$

where the fermions are taken in the fundamental (N -dimensional) representation of the group $SU(N)$ and

$$A_\mu = A_\mu^a T_a$$

with T_a being the $(N-1)$ mutually commuting generators of $SU(N)$ satisfying $[T_a, T_b] = 0$ and $\text{tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$.

For this case the loop average we have to compute is

$$W[C] = \frac{1}{N} \left\langle \text{tr} \left[\exp \left[iQ \oint_C A_\mu dx_\mu \right] \right] \right\rangle_{A, \bar{\psi}, \psi}. \quad (28)$$

Following exactly the same method we used for the $U(1)$ case, by using the effective Lagrangian for this model found in Ref. 9, and taking into account the color factorization in the path-integral averages and the T -matrices properties, we finally find that

$$\ln W_{\text{DQCD}_2}[C] = \frac{N-1}{2N} \ln W_{\text{QED}_2}[C], \quad (29)$$

where $\mu = e/\sqrt{\pi}$, the “photon” mass, must be replaced by $\mu = g/\sqrt{2\pi}$, the “gluon” mass,²² and the

factor $(N-1)/2N$ arises from

$$\sum_{i=1}^{N-1} T_{i(i+2)}^2 = \frac{N-1}{2N} I. \quad (30)$$

Therefore, all our conclusions referring to the Schwinger model trivially extend to massless DQCD₂.

III. CONCLUSIONS

We have computed the *exact* Wilson loop expectation value for QED₂ by using differently shaped contours. The generalization of these results to diagonal QCD₂ has also been discussed. By means of these results we have constructed *exact* quark-antiquark potentials.

Our results show that nonleading terms depend sensitively on the shape of the contour chosen in the loop computation. In particular, circular and elliptical contours give rise to long-range interactions between pairs that are known not to be present in the considered models.

On the other hand, it is known that rectangular contours have to be smooth in order to avoid

singularities⁷ in loop computations. But our exact result for an infinitely elongated ellipse shows that even at the first order in $1/d$, the shape of the contour used to round the corners becomes important.

Therefore, we see that one must be very careful when drawing physical conclusions from corrections to the leading term in the interquark potential, since these terms could be a pure artifact due to the way one is computing the loop.

ACKNOWLEDGMENTS

We would like to thank H. de Vega, K. Johnson, and H. Martin for useful discussions, C. G. Bollini and J. J. Giambiagi for valuable correspondence, and particularly Roman Jackiw for helpful discussions and for comments on the manuscript. F. S. wishes to thank the members of the Laboratoire de Physique Théorique et Hautes Energies de l'Université Pierre et Marie Curie, Paris, for their hospitality. This work was supported by CONICET and CIC, Argentina. The work of R. E. G. S. was supported in part through funds provided by the U. S. Department of Energy under Contract No. DE-AC020F6ER03069.

*On leave of absence from Universidad Nacional de La Plata, Argentina.

†Laboratoire associé au CNRS, France.

¹K. Wilson, Phys. Rev. D **10**, 2445 (1974).

²J. L. Gervais and A. Neveu, Phys. Lett. **80B**, 255 (1979); Y. Nambu, *ibid.* **80B**, 372 (1979); A. M. Polyakov, *ibid.* **82B**, 247 (1979); Yu Makeenko and A. A. Migdal, *ibid.* **88B**, 135 (1979).

³J. Schwinger, Phys. Rev. **128**, 2425 (1962); in *Theoretical Physics, Trieste Lectures, 1962* (I. A. E. A., Vienna, 1963), p. 89.

⁴Generalization of our results to diagonal QCD₂ will also be discussed.

⁵M. Abud, C. G. Bollini, and J. J. Giambiagi, Centro Brasileiro de Pesquisas Físicas, Report No. CBPF-NF-031/81, 1981 (unpublished).

⁶M. A. Shifman, Nucl. Phys. **B173**, 13 (1980).

⁷M. Luscher, K. Symanzik, and P. Weiz, Nucl. Phys. **B173**, 365 (1980); M. Luscher, *ibid.* **B180**, 317 (1981); J. Stack and M. Stone, Phys. Lett. **100B**, 476 (1981).

⁸A. Casher, J. Kogut, and L. Susskind, Phys. Rev. D **10**, 732 (1974).

⁹R. Gamboa Saraví, F. Schaposnik, and J. Solomin, Nucl. Phys. **B185**, 239 (1981).

¹⁰K. Fujikawa, Phys. Rev. Lett. **42**, 1195 (1979); Phys.

Rev. D **21**, 2848 (1980).

¹¹R. Roskies and F. Schaposnik, Phys. Rev. D **23**, 558 (1981).

¹²K. Johnson, Phys. Lett. **5**, 253 (1963). See also R. Jackiw, in *Laws of Hadronic Matter*, proceedings of the 1973 International Summer School "Ettore Majorana," Erice, Italy, edited by A. Zichichi (Academic, New York, 1975).

¹³It is worth noting that from dimensional considerations, $V(d) = Q^2 d F(ed)$ and $\lim_{d \rightarrow 0} F(ed) = \lim_{e \rightarrow 0} F(ed)$. Since taking $e \rightarrow 0$ corresponds to decoupling the quantum fermions from the "photon" — thus forbidding pair production and screening — both limits have to give the classical potential, $V(d) = (Q^2/2)d$.

¹⁴H. J. Rothe, K. D. Rothe, and J. A. Swieca, Phys. Rev. D **19**, 3020 (1979).

¹⁵Here T is an effective time, $T = \pi R$, $D = R$, and "lim" is to be understood as a limit taken when a factor T is separated first, leaving the rest of the R dependence on d .

¹⁶C. G. Bollini, J. J. Giambiagi, and J. Tiomno, Nuovo Cimento **59**, 412 (1980).

¹⁷R. Fukuda and Y. Kazama, Phys. Rev. Lett. **45**, 1142 (1980).

¹⁸L. Luscher, G. Münster, and P. Weiz, Nucl. Phys. **B180**, 1 (1981).

¹⁹Note that again we have defined T and d such that $Td = \pi ab$, thus $d = (\pi/2)b$ is the average distance between charges in a process that takes place in a time $T = 2a$.

²⁰I. V. Belvedere *et al.*, Nucl. Phys. **B153**, 112 (1979).

²¹P. J. Steinhardt, Harvard Report No. HUTP-80/A022, 1980 (unpublished).

²²Note that the factor $(2)^{-1/2}$ in the gluon mass arises from $\text{tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$.